

Product Requirements Document (PRD): HK Transit ETA (PWA)

Version: 2.3 (Master Compilation)

Status: Ready for Development

Platform: Progressive Web App (PWA) on Android

Development Stack: VS Code, TypeScript, React, Vite

Design System: Material You (Material Design 3)

1. Executive Summary

The goal is to develop a lightweight, high-performance Progressive Web App (PWA) that aggregates real-time arrival information (ETA) for major Hong Kong public transport operators.

Unlike standard apps that require repetitive searching, this product focuses on a "**Dashboard First**" approach, allowing users to group specific stops into contexts (e.g., "Morning Commute") and view all ETAs instantly.

2. Scope & Data Sources

2.1 Supported Operators

- **MTR** (Mass Transit Railway) - Urban Lines
- **Light Rail** (MTR Light Rail in NW New Territories)
- **KMB** (Kowloon Motor Bus)
- **LWB** (Long Win Bus)
- **CTB** (Citybus)
- **Joint Routes**: Cross-harbour routes operated jointly (KMB + CTB).

2.2 Core Value Propositions

1. **Unified Search**: One search bar for all operators.
2. **Instant List View**: Selecting a route shows ETAs for *all* stops immediately.
3. **Contextual Grouping**: Users can organize favorites into custom groups.
4. **Joint Route Logic**: Displaying combined departures (3 KMB + 3 CTB) for shared routes with robust filtering.

3. Functional Requirements

3.1 Module A: Search & Route Detail (The "List View")

- **FR-A1 (Unified Search)**: User can search by Route Number (e.g., "968", "101") or MTR Line Name.

- **FR-A2 (Immediate Detail):** Upon selecting a route, the app transitions to a **Scrollable List View** containing **ALL** stops for that route.
- **FR-A3 (Auto-Fetch ETA):** The app must automatically fetch and display the ETA for every stop in the visible list.
 - **Display Format:** 10min (10:01:10)
 - **Loading State:** Skeleton loader while fetching.
- **FR-A4 (Direction Toggle):** A prominent toggle (Segmented Button) at the top to switch between "Inbound" and "Outbound".

3.2 Module B: The Dashboard (Favorites & Groups)

- **FR-B1 (Long-Press to Save):** Long-press on any stop row -> "Add to Group" Bottom Sheet.
- **FR-B2 (Grouped Display):** Home Screen shows Group Cards (e.g., "Work", "Home").
- **FR-B3 (Batch Refresh):** Pull-to-refresh updates all; Group Header button updates specific group.
- **FR-B4 (Management):** Rename, delete, or reorder groups via drag-and-drop.

3.3 Module C: MTR & Light Rail Logic

- **FR-C1 (MTR):** Stations show ETAs for **both directions** simultaneously.
 - *Example:* "To Chai Wan: 2min (10:01:10) | To Kennedy Town: 4min (10:03:10)"
- **FR-C2 (Light Rail):** Ordered sequence of stops with ETAs.

3.4 Module D: Joint Route Logic (Updated)

- **FR-D1 (Identification Logic - "The 12A Exclusion"):**
The app must algorithmically determine if a route is a Cross-Harbour Joint Route using strict validation to avoid false positives (e.g., ensuring "12A" is treated as Local, while "101" is Joint).
 - **Rule:** A route is considered "Joint/Cross-Harbour" if and only if:
 1. It starts with 1, 3, 6, or 9.
 2. **AND** the numeric part of the route is ≥ 100 .
 - **Logic:**

```
const isJointRoute = (routeNo: string): boolean => {
  // Must start with 1, 3, 6, 9
  if (!/^(1|3|6|9)/.test(routeNo)) return false;

  // Parse the numeric part (e.g., "12A" -> 12, "101" -> 101, "962X" -> 962)
  const numericPart = parseInt(routeNo.replace(/D/g, ""), 10);

  // Exclude local routes like 12A, 3M, 6X which are < 100
  // Include cross-harbour routes like 101, 307, 601, 962 which are >= 100
  return numericPart >= 100;
};
```

- **FR-D2 (Dual Fetching):**
For identified Joint Routes, the app must fetch ETA data from BOTH KMB and CTB endpoints.
- **FR-D3 (Combined Display & Terminus Filtering):**
 - **Merge:** Display up to 6 departures (3 KMB + 3 CTB), sorted by time.
 - **Strict Direction Filtering (Terminus Fix):**
 - The app must strictly match the API response's dir (Direction/Bound) field with the currently selected route direction.
 - **Scenario:** If the user is viewing Route 101 **Inbound** (To Kwun Tong) and scrolls to the terminus (Kwun Tong), the app must **discard** any ETAs returned by the API that are labeled as **Outbound** (To Kennedy Town).
 - **Result:** The terminus stop should typically show "No Departure" or empty state, rather than showing the return trip's time incorrectly.

4. Technical Architecture (PWA Specification)

4.1 Technology Stack

- **Language:** TypeScript
- **Framework:** React + Vite
- **PWA:** vite-plugin-pwa
- **UI:** MUI (Material UI) v5+

4.2 Data Strategy

Data Layer	Storage	Purpose
Static	IndexedDB	Route lists, Stop names. Used to identify Joint Routes (check if Route No. exists in both KMB and CTB static JSONs).
Dynamic	React Query	Real-time ETA.

4.3 Batch Fetching & Filtering Logic (Detail)

- **KMB:** Use `/v1/transport/kmb/route-eta/{route}/{service_type}`.
- **Citybus:** Lazy load `/eta/{company_id}/{stop_id}/{route}`.
- **Joint Route Algorithm (FR-D1 & FR-D3 Implementation):**

```
// 1. Identification
if (isJointRoute(routeNo)) {
  // 2. Fetch Both
```

```

const [kmbData, ctbData] = await Promise.all([fetchKMB(), fetchCTB()]);

// 3. Filter by Direction (Crucial for Terminus)
// Assume currentViewDirection is 'O' (Outbound)
const validKMB = kmbData.filter(eta => eta.dir === 'O');
const validCTB = ctbData.filter(eta => eta.dir === 'O');

// 4. Merge & Sort
const combined = [...validKMB, ...validCTB]
  .sort((a, b) => a.timestamp - b.timestamp)
  .slice(0, 6); // Take top 6
}

```

5. UI/UX Design System (Material You)

5.1 Theming

- **Dynamic Color:** @material/material-color-utils based on seed color.

5.2 Component Specs

- **Route Lists (Standard):** 1-3 Lines of ETA.
- **Route Lists (Joint):**
 - **Format:** [LOGO] 10min (10:01:10)
 - **Visual:** Vertical list with small Operator Chips (Red for KMB, Yellow for CTB).

6. API Implementation Reference

A. KMB & LWB

- **Base:** <https://data.etabus.gov.hk>
- **Full Route ETA:** `/v1/transport/kmb/route-eta/{route}/{service_type}`
- **Note:** API returns dir as "I" (Inbound) or "O" (Outbound). Map this carefully.

B. Citybus (CTB)

- **Base:** <https://rt.data.gov.hk/v2/transport/citybus>
- **ETA:** `/eta/{company_id}/{stop_id}/{route}`
- **Note:** API returns dir as "I" or "O".

C. MTR

- **Base:** <https://rt.data.gov.hk/v1/transport/mtr/getSchedule.php>